Composite-clock phase selectors and an improved Möbius capacity floor

RH research program

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Abstract

The RH-366 distance-two capacity has an unconditional lower floor $4/\pi^2$ and a squarefree upper ceiling $6/\pi^2$, but its adaptive finite-prefix limit is open. We give a strict, source-locked improvement of the lower floor. A clocked selector with period $q = 180$ uses an explicit set of eighty phases having no two elements at distance two modulo 180. The resulting sign word is admissible for every possible Möbius input. Exact squarefree densities in the 180 arithmetic progressions and fixed-frequency Möbius cancellation give the correlation

$$\frac{97}{24\pi^2} = \frac{4}{\pi^2} + \frac{1}{24\pi^2}.$$

Consequently $\liminf K_N/N \geq 97/(24\pi^2)$. We also give a literal two-state universal-safety completion on the frozen four-state graph and check all 3240 input-pair rows. This is one explicit lower-bound witness, not an optimization over all clocks or memories; the ordinary capacity limit, intrinsic operators, prime-power traces, Riemann-zero identification, and RH remain outside scope.

1 Frozen capacity and claim boundary

The RH-366 scalar capacity is

$$K_N = \max_{\substack{\varepsilon_1,\ldots,\varepsilon_N \in \{-1,+1\} \\ \text{not } (\varepsilon_n,\varepsilon_{n+2})=(+1,+1)}} \left| \sum_{n \leq N} \mu(n)\varepsilon_n \right|. \tag{1}$$

The optimizer may read the complete prefix and may change with N. RH-366 proves

$$\frac{4}{\pi^2} \leq \liminf_{N\to\infty} \frac{K_N}{N} \leq \limsup_{N\to\infty} \frac{K_N}{N} \leq \frac{6}{\pi^2}, \tag{2}$$

without asserting convergence [4]. RH-371 reduces each finite value to eight odd-run frequencies but leaves their ordinary Möbius convergence open [3].

All statements here remain in the scalar adaptive-capacity data type. The selector is a prescribed arithmetic observable depending only on the phase and current Möbius value. It is not a canonical dynamical coupling, a trace, a determinant, or a model of Riemann zeros.

2 A general phase-selector lemma

Let $q \geq 1$ and let $I \subset \mathbb{Z}/q\mathbb{Z}$ satisfy

$$I \cap (I + 2) = \varnothing. \tag{3}$$

Define the one-site selector

$$e_r(a) = \begin{cases} +1, & r \in I \text{ and } a = +1, \\ -1, & \text{otherwise}, \end{cases} \qquad \varepsilon_n = e_{n \bmod q}(\mu(n)). \tag{4}$$

Lemma 2.1 (Universal admissibility)**.** *The sign word in* (4) *obeys the RH-366 distance-two rule for every ternary input word* $a_n \in \{-1, 0, +1\}$.

Proof. If $\varepsilon_n = +1$ and $\varepsilon_{n+2} = +1$, then both residues n and $n+2$ belong to I, contradicting (3). No arithmetic property of the input is used. $\square$

For a residue $r \pmod q$, write

$$\delta_{q,r} = \lim_{N\to\infty} \frac{1}{N} \#\{n \le N : n \equiv r \pmod q,\ \mu(n)^2 = 1\}. \tag{5}$$

Theorem 2.2 (Phase-selector correlation)**.** *For every finite* q *and* I *satisfying* (3),

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n\le N} \mu(n)\varepsilon_n = \sum_{r\in I} \delta_{q,r}. \tag{6}$$

Consequently

$$\sum_{r\in I} \delta_{q,r} \le \liminf_{N\to\infty} \frac{K_N}{N}. \tag{7}$$

Proof. Put $M_r(N) = \sum_{n\le N,\ n\equiv r\ (q)} \mu(n)$ and $Q_r(N) = \sum_{n\le N,\ n\equiv r\ (q)} \mu(n)^2$. On the selected residue classes,

$$\mu(n)e_r(\mu(n)) = -\mu(n) + 2\mathbf{1}_{\{\mu(n)=1\}} = -\mu(n) + \mu(n)^2 + \mu(n).$$

Thus the total sum equals

$$-\sum_{n\le N} \mu(n) + \sum_{r\in I} Q_r(N) + \sum_{r\in I} M_r(N).$$

Fixed-frequency Davenport cancellation, followed by finite Fourier inversion in the fixed arithmetic progressions, gives $M_r(N) = o(N)$, and the squarefree progression sieve gives $Q_r(N)/N \to \delta_{q,r}$ [1, 2]. The global Möbius sum is also $o(N)$, proving (6). The lemma makes the selector admissible, so its absolute correlation is bounded by K_N for every N; taking the liminf gives (7). $\square$

Remark 2.3 (No optimization claim). The theorem applies to every fixed finite clock, but it does not say that the best I for a given q has been found, nor that the supremum over q equals the adaptive capacity limit. A finite selector is a lower certificate only.

3 The composite clock $q = 180$

Take $q = 180 = 2^2\, 3^2\, 5$. The squarefree progression density has the exact piecewise form

$$\pi^2 \delta_{180,r} = \begin{cases} 0, & 4 \mid r \text{ or } 9 \mid r, \\ 5/96, & 4 \nmid r,\ 9 \nmid r,\ 5 \nmid r, \\ 1/24, & 4 \nmid r,\ 9 \nmid r,\ 5 \mid r. \end{cases} \tag{8}$$

Derivation of (8)*.* The congruence class is empty of squarefree integers when it is forced to be zero modulo 4 or 9. Otherwise, the unrestricted primes contribute

$$\frac{1}{180} \prod_{p\nmid 180} (1 - p^{-2}) = \frac{1}{180} \frac{6/\pi^2}{(1-2^{-2})(1-3^{-2})(1-5^{-2})} = \frac{5}{96\pi^2}.$$

When $5 \mid r$, the class is zero modulo 5 and one of the five lifts modulo 25 is forbidden, multiplying by $1 - 1/5 = 4/5$ and giving $1/(24\pi^2)$. $\square$

Use the following explicit phase set:

$$\begin{aligned}
I_{\text{even}} &= \{r \in \mathbb{Z}/180\mathbb{Z} : r \equiv 2 \pmod 4,\ 9 \nmid r\},\\
I_{\text{odd}} &= \{3, 7, 11, 15, 19, 23, 29, 33, 37, 41, 47, 51, 57, 61, 65, 69,\\
&\qquad 73, 77, 83, 87, 91, 97, 101, 105, 109, 113, 119, 123, 127, 131,\\
&\qquad 137, 141, 147, 151, 155, 159, 163, 167, 173, 177\},\\
I &= I_{\text{even}} \cup I_{\text{odd}}.
\end{aligned}$$

The even set has 40 elements and the displayed odd list has 40 elements. Direct reduction modulo 180 gives

$$I \cap (I + 2) = \varnothing. \tag{9}$$

Among the selected phases, 68 have density coefficient 5/96 and 12 have coefficient 1/24; none has coefficient zero. Therefore

$$\sum_{r \in I} \delta_{180,r} = \frac{68 \cdot 5 + 12 \cdot 4}{96\pi^2} = \frac{97}{24\pi^2}. \tag{10}$$

Theorem 3.1 (Improved RH-366 lower floor)**.** *For the RH-366 distance-two capacity,*

$$\boxed{\liminf_{N \to \infty} \frac{K_N}{N} \geq \frac{97}{24\pi^2}} \tag{11}$$

and hence

$$\frac{97}{24\pi^2} = \frac{4}{\pi^2} + \frac{1}{24\pi^2} > \frac{4}{\pi^2}.$$

Proof. Apply Theorem 2.2 to the set I in (9) and use (10). ☐

4 Literal graph completion

For completeness, the selector can be represented by a universally safe finite-memory table on the frozen RH-366 graph. Its state set is $\{M, P\}$, and the graph vertices are ordered as

$$(--), \quad (-+), \quad (+-), \quad (++),$$

with observable $(-1, -1, +1, +1)$ and the six frozen edges from RH-366. For $r \in \mathbb{Z}/180\mathbb{Z}$ put

$$b_M(r) = -1, \qquad b_P(r) = \begin{cases} +1, & r - 1 \in I, \\ -1, & r - 1 \notin I, \end{cases}$$

and let

$$e_r(a) = \begin{cases} +1, & r \in I,\ a = +1, \\ -1, & \text{otherwise.} \end{cases}$$

The output in state s is the pair $(e_r(a), b_s(r))$, and the next state is P exactly when $e_r(a) = +1$ (otherwise it is M).

Proposition 4.1 (Universal-safety completion)**.** *The table above is universally safe and has the one-site observable $e_r(a)$.*

Proof. If $e_r(a) = -1$, the next state is M and its previous-sign coordinate at phase $r + 1$ is $-1 = e_r(a)$. If $e_r(a) = +1$, then $r \in I$, the next state is P, and $b_P(r + 1) = +1 = e_r(a)$. Thus the pair coordinates shift correctly. The only forbidden edge would have current previous sign $+1$ and next current sign $+1$. The former implies $r - 1 \in I$ and the latter implies $r + 1 \in I$, contradicting $I \cap (I + 2) = \varnothing$. The first coordinate of every output is $e_r(a)$, so the one-site condition is literal for both states. ☐

This completion is a finite certificate, not a new operator. It merely embeds the explicit sign word in the already frozen graph language.

5 Executable audit

The exact artifact uses an integer Möbius sieve, rational density coefficients, and the table construction above. Its principal rows are:

selected phases	80
distance-two conflicts	0
coefficient-5/96 phases	68
coefficient-1/24 phases	12
universal state/phase/input-pair rows	3240
prefix witness rows ($1 \leq N \leq 2048$)	2048
endpoint	$N = 2^{16}$
endpoint selector score	26852
endpoint capacity	32320

The endpoint rows and all prefix rows are finite reproducibility checks, not asymptotic evidence. Source hashes for RH-366, RH-371, RH-372, the arithmetic source, and the four-volume archive are stored in `results/result.json`.

6 Route verdict and Gate ledger

Route A is `GO` narrowly: the phase-selector theorem and the explicit $q = 180$ density calculation are unconditional and independent. Route B is `STOP_SCOPED`. This paper does not prove convergence of K_N/N, does not optimize over all clocks or memory budgets, and does not supply the higher-order Möbius correlations needed for the unrestricted genuinely memory-dependent class.

Gates A–E remain false/open. In particular, the selector uses an externally prescribed phase and current Möbius input and has no canonical operator, Fredholm determinant, von-Mangoldt prime-power trace, self-adjoint generator, zero identification, Hilbert–Polya construction, or implication for the Riemann Hypothesis.

7 Conclusion

A composite clock can exploit squarefree residue-class defects without violating the distance-two sign rule. The explicit $q = 180$ witness raises the rigorous capacity floor by $1/(24\pi^2)$. The remaining gap to the squarefree ceiling is still a nonlinear adaptive problem: the new certificate does not turn a finite selector family into an asymptotic capacity theorem. The next mathematical edge must address that distinction rather than extrapolate from the finite endpoint.

References

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[3] RH research program. Eight-run reduction and the cyclic pair-ledger obstruction, 2026. Repository paper RH-371, frozen release.

[4] RH research program. Mobius orthogonality, adaptive encoding, and parry covariance, 2026. Repository paper RH-366, frozen release.